

Project Initialization and Planning Phase

Date	20 June 2026
Team ID	
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform flood disaster management and early warning mechanisms using machine learning, boosting prediction efficiency and environmental analytical accuracy. It tackles system inefficiencies found in traditional monitoring frameworks, promising better preparation operations, reduced regional hazards, and improved community safety levels. Key features include a high-accuracy machine learning-based prediction model and real-time decision-making support via web interactions.

Project Overview	
Objective	The primary objective is to revolutionize the disaster preparedness process by implementing advanced machine learning techniques, ensuring faster, more accurate regional flood probability assessments.
Scope	The project comprehensively assesses and enhances the flood monitoring pipeline, incorporating machine learning algorithms for a more robust, automated, and efficient early warning web platform.
Problem Statement	
Description	Addressing inaccuracies and long computation delays in current baseline weather monitoring systems adversely affects municipal operational efficiency and proactive community evacuation safety.
Impact	Solving these issues will result in improved local operational readiness, reduced economic risks to property, and an overall enhancement in disaster mitigation, contributing to regional safety stability and structural organizational success.
Proposed Solution	
Approach	Employing machine learning classification techniques to analyze and predict environmental vulnerability factors, creating a dynamic, web-adaptable flood status response system.
Key Features	Implementation of a machine learning-based predictive evaluation model utilizing the high-performing XGBoost algorithm.

	Real-time decision-making for quicker status assessments via a Flask web application layer. Feature engineering and automated preprocessing objects to cleanly adapt incoming weather patterns.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Multi-core Central Processor (Intel i5/AMD Ryzen 5 equivalent or cloud runtime environment)
Memory	RAM specifications	8 GB or higher
Storage	Disk space for data, models, and logs	1 TB SSD or 256 GB SSD (Local Development & Model storage space)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost, joblib
Development Environment	IDE	Anaconda Navigator, Jupyter Notebook, VS Code, or PyCharm
Data		
Data	Source, size, format	Kaggle open-source dataset, multi-column feature framework (Annual rainfall, Cloud visibility, Seasonal rainfall), .csv format